

Course Syllabus

BIS 372 Business Systems Design and Development

Prerequisites: BA 272 [C-] and (ACTG 378(H) or BA 370(H) [C-]) and (BIS 371 or BA 371 [C-])

4 Credits

Andy Cadotte

College of Business

Email: cadottea@oregonstate.edu (<https://canvas.oregonstate.edu/mailto:cadottea@oregonstate.edu>)

Office hours

5:30pm - 6:30pm pst Tuesday and Thursday

<https://oregonstate.zoom.us/j/98682916911?pwd=ebezS6zpEcpmlH7vyJvm16eCOFCAp.1>

[➡ \(https://oregonstate.zoom.us/j/98062533981?pwd=y3Ri1XRWIZBR0cUoIRaAeaZZEoVBHd.1\)](https://oregonstate.zoom.us/j/98062533981?pwd=y3Ri1XRWIZBR0cUoIRaAeaZZEoVBHd.1)

Course Description (OSU Course Catalog)

Explores how information systems are developed, deployed, and adapted considering architectures, value propositions, infrastructure, and organizational factors. Explores both classic and agile application design and development approaches and tasks, tools and methods of system developers and automation tool users.

Course Description (Extended)

In BIS 372 we concentrate on IS analysis, design and development (D&D). We address general systems D&D principles, classic IS architectures with an emphasis on web architectures, and different software development approaches with an emphasis on agile methods. We also explore the terminology-filled world and culture of system developers, discuss trends in system development and the ever evolving development-test-release/deploy cycle and discuss and explore open source technologies and processes. We will practice some of the technologies we

discuss as part of this class. As such, we will rely and expand on the programming skills we developed in BA 272. We will add some scripting (Linux shell, Python, JavaScript) and practice a Python web technology.

Course Learning Outcomes

Upon completion of this course you should be able to:

- Specify and evaluate conceptual architectures for business information systems solutions.
- Research design solutions for IS architectural problems.
- Design basic IS architectures and propose components and relationships for those architectures.
- Communicate and explain design proposals and decisions within and across teams of designers or a sponsoring agency.
- Architect and prototype part of a real-world system in an architecture of your choice.
- Work in and contribute to a system design and development team.
- Understand, explain and apply the concepts and practices of agile development.
- Communicate about and feel somewhat comfortable with Linux (read: Unix-like) machines.
- Develop simple Powershell scripts.
- Understand how to live with and overcome the statelessness of HTTP.
- Develop basic web (HTTP)-based programs (2-4 tier).

Course Components

- Lectures and texts.
- Self-guided lab exercises needed to successfully complete assignments.

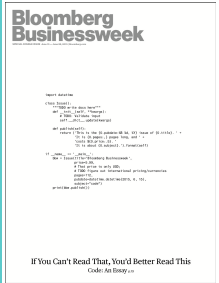
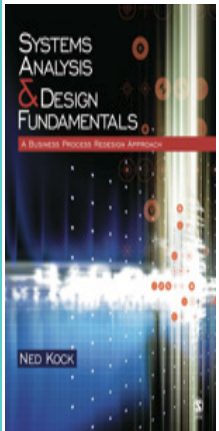
Note: The lab exercises have been carefully designed to fit the assignments in this course.

- Three (3) coding assignments.
- Since code reviews are a normal part of IS systems work, you will each be asked to review the coding assignments of two of your colleagues.
- **Class project** (<https://canvas.oregonstate.edu/courses/2055838/pages/class-project>): teams of

up to 5 people (or you, working by yourself) will each design and implement a small Design & Development (D&D) project.

- Two (2) midterm exams (graded -- essay questions; answer n from m questions).

Texts

Short, online texts	See Texts in Course Schedule below
	Ford, P. (2015) The Code Issue. <i>Bloomberg Business Week Magazine</i>. Freely available on-line through OSU-library: 1. Point web browser to OSU library (https://library.oregonstate.edu/) 2. Select <i>Databases A - Z</i> 3. Select <i>Business Source Complete (EBSCOhost)</i> 4. Search for "<i>the code issue</i>" 5. Select <i>PDF Full Text</i> 6. Magazine's table of content is listed on the left; by groups of pages; each section with its own link. This issue of Bloomberg Business Week magazine was published in June 2015 and introduces business managers to the world of software development and software developers. The magazine is divided into sections and each week we will read and discuss some of these sections (See the links in The Code Issue column https://cob-te-web.business.oregonstate.edu/public_html/peer_eval/peer_eval.html in Course Schedule table below).
	Kock, N. (2007) <i>Systems Analysis & Design Fundamentals. A Business Process Redesign Approach</i>. Sage Publications. (Freely available on-line through OSU library/ONID https://ebookcentral.proquest.com/lib/osu/detail.action?docID=996263). (Selected chapters)

Course Schedule

(Items due, deliveries, exams, etc. in italics)

Timing	Theory/Lectures Instructor Notes	The Code Issue; instructor notes (For recorded discussion of these notes, see the associated entries in the weekly Modules)	Labs, Assignments, Exam
Week 1			
	Welcome & syllabus SDLC (https:// canvas.oregonstate.edu/ courses/2055838/pages/ introduction-system- slash-software- development-life-cycle- sdlc)	Cover page (https:// classes.business.oregonstate.edu/ ba372/ the_code_issue.html#the_code) TMitTB (https:// classes.business.oregonstate.edu/ ba372/ the_code_issue.html#the_man) Why are we here? (https:// classes.business.oregonstate.edu/ ba372/ the_code_issue.html#why_here)	Self-directed lab: Python refresher (https:// canvas.oregonstate.edu/ courses/2055838/pages/python refresher-and-a-few-new-things Self-directed lab: Python GU with Tkinter (https:// canvas.oregonstate.edu/ courses/2055838/pages/python simple-guis-with-tkinter)
Week 2			
	HTTP (https:// canvas.oregonstate.edu/ courses/2055838/pages/ the-web-http)	Let's begin (https:// classes.business.oregonstate.edu/ ba372/the_code_issue.html#begin) How Do You Type an A? (https:// classes.business.oregonstate.edu/ ba372/the_code_issue.html#type_a) From Hardware to Software (https:// classes.business.oregonstate.edu/ ba372/the_code_issue.html#hw_sw)	Self-directed lab: Python DB interaction (https:// canvas.oregonstate.edu/ courses/2055838/pages/python database-communications)

		<u>How Does Code Become Software?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#code2sw) <u>What is an Algorithm?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#algorithm)	
Week 3			
	<u>Business Process Improvement</u> (https://canvas.oregonstate.edu/courses/2055838/modules/items/26807795)	<u>The Sprint</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#sprint) <u>What's With All These Conferences?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#confs) <u>Why Are Programmers So Intense About Languages?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#intense) <u>Why Are There So Many Languages?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#languages) <u>The Beauty of the Standard Library</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#standard)	<u>Assignment 1 due</u> (https://canvas.oregonstate.edu/courses/2055838/assignments/10525907)
Week 4			

	<u>IS impact assessment</u> (https://canvas.oregonstate.edu/courses/2055838/pages/is-impact-assessment)	<u>What Do Different Languages Do?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#diff_languages) <u>The Importance of C</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#c) <u>The Corporate Object Revolution</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#corporate) <u>Look How Big and Weird Things Get With Just Python</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#python)	<u>Self-directed lab: Python -- OOP</u> (https://canvas.oregonstate.edu/courses/2055838/pages/python-just-a-little-oop) <u>Assignment 1 code reviews due</u> (https://canvas.oregonstate.edu/courses/2055838/pages/instructions-for-submitting-and-grading-assignments) <i>Midterm exam 1</i>
Week 5			
	<u>IS Architecture</u> (https://canvas.oregonstate.edu/courses/2055838/modules/items/26807804) <u>Compilation -- Interpretation --Byte-compilation</u> (https://canvas.oregonstate.edu/courses/2055838/modules/items/26807805)	<u>What About JavaScript?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#javascript) <u>Why are Coders Angry?</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#angry) <u>The Legend of the 10x Programmer</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#10x) <u>The Thing With Real Artists Is That They...</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#artists) <u>We Still Need to Choose...</u> (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#choose)	<u>Self-directed lab: Python -- JSON</u> (https://canvas.oregonstate.edu/courses/2055838/pages/python-json) <u>JSON</u> (https://classes.business.oregonstate.edu/ba372/python_json.html)

		classes.business.oregonstate.edu/ba372/the_code_issue.html#choose)	
Week 6			
	Usability (UX) (https://canvas.oregonstate.edu/courses/2055838/modules/items/26807809)	The Time You Attended the Email Address Validation Meeting (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#validation) The language of White Collars (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#white_collars)	Assignment 2 due (https://canvas.oregonstate.edu/courses/2055838/assignments/10525908) Self-directed lab: Python — Flask (https://canvas.oregonstate.edu/courses/2055838/pages/flask-python-on-the-web)
Week 7			

		Liquid Infrastructure (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#liquid) Off the Shelf (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#shelf) What's the Absolute Minimum I Must Know About PHP? (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#php)	Assignment 2 code reviews due (https://canvas.oregonstate.edu/courses/2055838/pages/instructions-for-submitting-and-grading-assignments) Self-directed lab: Powershel command line lab (https://canvas.oregonstate.edu/courses/2055838/pages/practic/lab-powershell-command-line-l) Self-directed lab: Powershel scripting lab (https://canvas.oregonstate.edu/courses/2055838/pages/practic/lab-powershell-scripting-lab)
Week 8			
	FOSS (https://canvas.oregonstate.edu/courses/2055838/pages/free-and-open-source-software-foss)	How Are Apps Made? (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#apps) The Framework: Wilder, Younger Cousin of the Software Development Kit (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#frameworks) What is Debugging? (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#debugging) Nothing Is Built (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#nothing)	<i>Midterm exam 2</i>

		How Does Testing Work? (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#testing) And Now For Something Beautiful (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#beautiful)	
Week 9			
		The Triumph of Middle Management (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#triumph) How Do You Pick a Programming Language? (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#pick) Welcome to the Scrum (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#welcome) Managing Programmers (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#mgmt) We Are Going to Ship (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#ship) Should You Learn to Code? (https://classes.business.oregonstate.edu/ba372/the_code_issue.html#code)	Assignment 3 due (https://canvas.oregonstate.edu/courses/2055838/assignments/10525909)

Week 10			
			<u>Assignment 3 code reviews due (https:// canvas.oregonstate.edu/ courses/2055838/pages/ instructions-for-submitting-and grading-assignments)</u>

Grading

Course grading

Description	Points	Points
	Individual	Team
Class project: Delivery 1	20	20
Class project: Delivery 2	20	15
Class project: Peer evaluation grade (teams only; see notes below)	NA	5
Midterm exam 1	15	15
Midterm exam 2	15	15
Assignments (see notes below)	60	60
Self-Directed Labs	40	40
Total:	170	175

Notes on grading

- **Class Project:** To receive peer evaluation credit, you must yourself turn in the evaluation of your peers.

- **Assignments:** You will only receive credit for your coding work if you turn in good, serious and on-time code reviews.
- **Team Grade:** 'Team grade' means that all members of the team receive the same grade. However, as an individual, you will only receive that team grade if you have appropriately and sufficiently contributed to the team's accomplishments. Simply put: if you do not contribute, you will not receive team credit. Also, your instructor may ask you, as a team, to redo a class project report. Such redo's, however, suffer a 15% grade penalty.

Number-to-letter grading scale

The following number-to-letter-grade scale will be used for calculating the final course letter grade. The minimum percentage of available points to earn the applicable grade is:

Grading scale (minimum percentages)

A	93%	C	73%
A-	90%	C-	70%
B+	87%	D+	67%
B	83%	D	63%
B-	80%	D-	60%
C+	77%	F	<60%

Course Policies

Student conduct is governed by the university's policies as explained in the Student Conduct Code (<https://beav.es/codeofconduct> (<https://beav.es/codeofconduct>)). Students are expected to conduct themselves in the course; *e.g.*, on discussion boards, in email postings, in team project work. *etc.*, in compliance with the university's regulations regarding civility.

0. COB Code of Honor



A code of honor represents the moral commitments of those abiding to it. While each person lives by his or her personal code, the establishment of collective values creates a universal goal to which we can aspire. It is through the pursuit of these professional attributes that we reduce the possibility of immoral actions ourselves. In order to uphold our personal character and the organization that we proudly call our own, we take this oath.

- **Integrity:** *The quality of being honest and having strong moral principles, integrity stands as the backbone of character and is essential for success.*
- **Respect:** *Respect for others and yourself is a commitment to the fair treatment of and the fair competition with others. Through respect we embolden the character of others and ourselves.*
- **Responsibility:** *We are held accountable for our words and actions as professionals to embed a steadfast commitment to honor in our decisions.*

1. Academic Misconduct

According to OSU student conduct regulations, academic misconduct is defined as an intentional act of deception in which a student seeks to claim credit for the work or effort of another person or uses unauthorized materials or fabricated information in any academic work, either through the student's own efforts or the efforts of another. Refer to the **OSU Code of Student Conduct** (<https://studentlife.oregonstate.edu/pre-student-conduct-community-standards>) for a comprehensive definition of academic misconduct. All cases of suspect academic misconduct will be handled in strict accordance with **OSU policy** (<http://studentlife.oregonstate.edu/studentconduct>) and **College of Business Academic and Professional Standards** (<https://business.oregonstate.edu/student-experience/advising/resources/academic-policy-standards>).

2. A Word on AI (ChatGPT, etc.)

Much is being written/said about ChatGPT and other AIs these days. For the purpose of this class consider ChatGPT and similar products as information sources but do not ask them to do your work. With that we mean that if you are asked to write an assignment or a text, you do not ask ChatGPT to do it for you as that would constitute a clear act of plagiarism/fraud; *i.e.*, pretending that ChatGPT's work is your work.

3. Digital Hygiene

Please, keep your digital devices clean and updated. Make a habit of regularly —we suggest weekly— updating your devices so that you and your on-line communities stay safe and so that IT support services can actually support you.

4. Title 9

OSU is committed to providing a learning environment that is free of all forms of sexual

discrimination and misconduct. Please be aware that your instructor must report incidents you disclose to me inside or outside the classroom that involve gender or sex-based harassment, violence, or discrimination, including your name, to the Office of Equal Opportunity and Access (EOA). For more information on how EOA responds to reports, please visit their website at eo.oregonstate.edu (<https://eo.oregonstate.edu/>). However, if you wish to make a confidential disclosure and receive information on resources and services, please contact the Survivor Advocacy and Resource Center (SARC) by phone (541-737-2030), by e-mail (survivoradvocacy@oregonstate.edu (<mailto:survivoradvocacy@oregonstate.edu>)), or visit them in the Plageman Building.

Students with Disabilities

Accommodations for students with disabilities are determined and approved by Disability Access Services (DAS). If you, as a student, believe you are eligible for accommodations but have not obtained approval, please contact DAS immediately at 541-737-4098 or at <http://ds.oregonstate.edu> (<http://ds.oregonstate.edu>). DAS notifies students and faculty members of approved academic accommodations and coordinates implementation of those accommodations. While not required, students and faculty members are encouraged to discuss details of the implementation of individual accommodations.

Instructor Commitment to Equity, Diversity and Inclusion

As your instructor I believe that the diverse backgrounds and perspectives of students and the larger university community are a strength and a benefit, both in class and in society. I strive to reflect and enact OSU's commitment to embracing equity, diversity and inclusion in all aspects of my teaching, research and service at OSU. You should be aware that some of the materials we cover may be difficult to read and/or discuss, or may represent opinions that you find offensive or objectionable. As we examine this material, it is essential that we are respectful to each other both online and face-to-face.

Your suggestions for improving this course are encouraged and appreciated, and I hope that you will contact me if you have any questions or concerns. Each of us has a responsibility to work to create an environment free of harassment and discrimination. Please take a moment to read the [University Policy on discrimination and harassment](https://registrar.oregonstate.edu/osu-discrimination-and-harassment-policies) (<https://registrar.oregonstate.edu/osu-discrimination-and-harassment-policies>). If you feel you have experienced sexual misconduct, discrimination, harassment, bullying, or retaliation, please see the [Office of Equal Opportunity and Access](https://eo.oregonstate.edu) (<https://eo.oregonstate.edu>) for information about reporting.

Reach Out for Success

University students encounter setbacks from time to time. If you encounter difficulties and need assistance, it is important to reach out. Consider discussing the situation with your instructor or academic advisor and learn about [resources that assist with wellness and academic success \(https://counseling.oregonstate.edu/reach-out-success\)](https://counseling.oregonstate.edu/reach-out-success).

For mental health:

- If you are in immediate crisis, please contact the Crisis Text Line by texting *OREGON* to 741-741 or call the National Suicide Prevention Lifeline at 1-800-273-TALK (8255).

For financial hardship:

- Any student whose academic performance is impacted due to financial stress or the inability to afford groceries, housing, and other necessities for any reason is urged to contact the Director of Care for support (541-737-8748).

Student Bill of Rights

OSU has twelve established student rights. They include due process in all university disciplinary processes, an equal opportunity to learn, and grading in accordance with the course syllabus: [https://asosu.oregonstate.edu/advocacy/rights \(https://asosu.oregonstate.edu/advocacy/rights\)](https://asosu.oregonstate.edu/advocacy/rights)

Academic Calendar

All students are subject to the registration and refund deadlines as stated in the [Academic Calendar \(https://registrar.oregonstate.edu/osu-academic-calendar\)](https://registrar.oregonstate.edu/osu-academic-calendar).